Sheaves and Cohomology Workbook

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Remark

Problem. Denote by $\mathrm{Mod}^f(A)$ the full subcategory of $\mathrm{Mod}(A)$ consisting of finitely generated modules. Does the category $\mathrm{Mod}^f(A)$ admit kernels, cokernels, finite/infinite direct sums, products?

Solution.

- **Kernels.** Kernel of the morphism $N \xrightarrow{\phi} M$ between two finitely generated modules is not always finitely generated. Take $A = N = \mathbf{k}[x_1, x_2, \dots]$ and $M := \mathbf{k}[x_1, x_2, \dots]/I$ where $I = (x_1, x_2, \dots)$. We have morphism $A \to A/I$ which has a kernel I. Surely, I is not finitely generated as an A-module, but $A/I \cong \mathbf{k}$ is finitely generated.
- **Cokernels.** Assume $N \xrightarrow{\psi} M$ is another morphism of f.g. modules. Is $M/\mathrm{im}(\psi)$ a finitely generated module? Yes. Note that $\mathrm{im}(\psi)$ is finitely generated by $\psi(n_1), \dots, \psi(n_r)$ where $N = (n_1, \dots, n_r)$. Quotient of two finitely generated modules is finitely generated.
- **Finite direct sums.** Obviously $\bigoplus_{i \in I} N_i$ for finite I is finitely generated by the elements $e_{ij} = (0, 0, \dots, n_{ij}, \dots, 0)$ where $i \in I$ and j ranges over generators of N_i.
- **Infinite direct sums.** For instance, $\mathbb{Z}^{\oplus \mathbb{N}}$ is not finitely generated.
- **Finite products.** Since finite products are the same as finite direct sums in $\mathrm{Mod}(A)$, we have already covered this case.
- **Infinite products.** Take $\prod_{i \in \mathbb{N}} \mathbb{Z}$. It is not finitely generated.

Conclusion: $\mathrm{Mod}^f(A)$ only admits cokernels, finite direct sums and products.

Problem. Let X be a set and $\mathscr{C}$ be a category $\mathscr{P}(X)$ of all subsets of X, the morphism being the inclusion morphism. Does $\mathscr{C}$ admit an initial object, terminal object, products, coproducts?

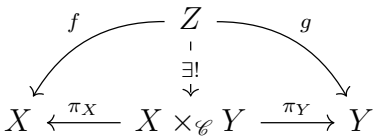
Solution. $P \in \mathscr{C}$ is initial if $\mathrm{Hom}_{\mathscr{C}}(P, X) \cong \{\mathsf{pt}\}$ for all $X \in \mathscr{C}$. By definition recall:

$$\mathrm{Hom}_{\mathscr{C}}(P, X) = \begin{cases} \{\mathsf{pt}\} & P \subseteq X, \\ \emptyset & \text{otherwise.} \end{cases}$$

Thus $P \subseteq X$ for any $X \in \mathscr{P}(X)$. This is the case when $P = \emptyset$.

P is terminal if $\mathrm{Hom}_{\mathscr{C}}(X, P) \cong \{\mathsf{pt}\}$ for all $X \in \mathscr{P}(X)$. This is equivalent to $X \subseteq P$ for all $X \in \mathscr{P}(X)$. This is the case when $P = X$.

How to define the product $X \times Y$? By definition, the following diagram should commute:



So naturally $X \times_{\mathscr{C}} Y = X \cap Y$. Similarly, $X \oplus_{\mathscr{C}} Y = X \cup Y$. Obviously, $\bigcap_{i \in I} X_i$ and $\bigcup_{i \in I} X_i$ are always defined in $\mathscr{P}(X)$ for any cardinality of I.

Problem. Let X be a topological space and let $b \in X$. One defines a presheaf F on X by setting $F(U) = 0$ if $b \notin U$ and $F(U) = \mathbf{k}$ if $b \in U$. Prove that F is a sheaf and give a short description of it by using the direct image functor.

Solution. F is a skyscrapper sheaf δ_b that is defined as follows: let $i_b : \{b\} \hookrightarrow X$ be the embedding into X, then $F \cong i_{b*} i_b^{-1} \mathbf{k}_X$. By definition, $f^{-1}F$ is a map that associates a sheaf to a presheaf $f^\dagger F$, thus $i_b^{-1}\mathbf{k}_X$ is a sheaf. Since f_* sends sheaf to a sheaf, the composition yields a sheaf.

Problem. Let X be an open subset of $\mathbb{C}$ and let $b \in X$. For U open in X, one denotes by $\mathscr{O}_X^b(U)$ the subspace of $\mathscr{O}_X(U)$ consisting of holomorphic functions vanishing at b (Hence, if $b \notin U$, $\mathscr{O}_X^b(U) = \mathscr{O}_X(U)$). Show that $U \mapsto \mathscr{O}_X^b(U)$ is a sheaf, that we denote by $\mathscr{O}_X^b$ and describe the sheaf F^b given by the exact sequence of sheaves $0 \to \mathscr{O}_X^b \to \mathscr{O}_X \to F^b \to 0$.

Solution. By definition $\mathscr{O}_X^b(U) = \{f \in \mathscr{O}_X(U) : f(b) = 0\}$. Let us check sheaf conditions:

- *Locality.* Obvious.
- *Glueing.* Suppose $\{U_i\}_{i \in U}$ is an open cover of U, and family $\{s_i \in \mathscr{O}_X^b(U_i)\}_{i \in I}$ satisfies the property $s_i|_{U_{ij}} = s_j|_{U_{ij}}$ for all i, j. By the property of $\mathscr{O}_X$, there exists some $s \in \mathscr{O}_X$ such that $s|_{U_i} = s_i$. We shall check that $s \in \mathscr{O}_X^b$, that is $s(b) = 0$ (of course assume $b \in U$, otherwise the statement is obvious). But then there exists some U_i such that $s|_{U_i}(b) = 0$, meaning $s(b) = 0$.

Now consider the exact sequence

$$0 \to \mathscr{O}_X^b \to \mathscr{O}_X \to F^b \to 0.$$

By the definition of the exact sequence, $F^b = \mathrm{Coker}(\mathscr{O}_X^b \to \mathscr{O}_X)$. Take some open U and by definition of the cokernel, F^b is a sheafication of $U \mapsto \mathscr{O}_X(U)/\mathscr{O}_X^b(U)$. If $b \notin U$, then this corresponds to a constant sheaf $U \mapsto 0$. Otherwise, this quotient defines the equivalence classes of functions that agree on b, thus this is isomorphic to $\mathbb{C}$. Therefore, we have the skyscrapper sheaf $F^b = i_{b*} i_b^{-1} \mathbb{C}_X$.

Problem. Let $X = \mathbb{R}$ and consider the morphism of sheaves $P : C_X^\infty \to C_X^\infty$, where P is the differential operator $\partial_x - 1$.

(i) Calculate $\mathrm{Ker}(P)$ and $\mathrm{Coker}(P)$.

(ii) Same question with P replaced with $Q = x\partial_x - \frac{1}{2}$.

Solution. *Part (i).* By definition, $\mathrm{Ker}(P) = \{f \in C_X^\infty : P(f) = 0\}$. Thus we obtain $f' - f = 0$ or $f = \gamma e^x$ for $\gamma \in \mathbb{R}$. Thus, $\mathrm{Ker}(P) = \mathbb{R}e^x$. Now recall $\mathrm{Coker}(P) = C_X^\infty/\mathrm{Im}(P)$. So what is $\mathrm{Im}(P)$? Note $g \in \mathrm{Im}(P)$ iff $g = f' - f$ has a solution w.r.t f. But this is always the case, so $\mathrm{Im}(P) = C_X^\infty$ and so $\mathrm{Coker}(P) = 0$.

Part (ii). Again, $\mathrm{Ker}(P) = \{f \in C_X^\infty : P(f) = 0\}$. Thus we have $xf'(x) - \frac{1}{2}f(x) = 0$ or $f(x) = c\sqrt{|x|}$. But note that $f(x) \notin C_X^\infty$. Thus, $\mathrm{Ker}(P) = 0$.

Now consider $\mathrm{Im}(P)$, so we have the equation

$$x\frac{df}{dx} - \frac{1}{2}f(x) = g(x)$$

One can verify that:

$$f(x) = \sqrt{|x|}\left(\gamma + \int_0^x \frac{g(t)dt}{t\sqrt{|t|}}\right), \quad x \neq 0$$

So whenever we have the equation on some open $U \not\ni 0$, then we always have a solution. Near 0, however, there are some non-trivial solutions, which we cannot describe. Thus, $\mathrm{Coker}(P) \cong i_*\mathbb{R}_{\{0\}}$.

Problem. Let $M, N \in \underline{\mathrm{Mod}}(\mathbf{k})$ and consider the functor $\theta : \underline{\mathrm{Mod}}(\mathbf{k}) \to \underline{\mathrm{Mod}}(\mathbf{k})$ given by $L \mapsto \mathrm{Hom}_{\mathbf{k}}(M \otimes_{\mathbf{k}} L, N)$. Is this functor covariant/contravariant? Left/right exact?

Solution. We shall think how to map morphisms $f \in \mathrm{Hom}_{\mathbf{k}}(L, R)$ to $\theta(f) : \mathrm{Hom}_{\mathbf{k}}(M \otimes_{\mathbf{k}} R, N) \to \mathrm{Hom}_{\mathbf{k}}(M \otimes_{\mathbf{k}} L, N)$. Recall that

$$\mathrm{Hom}_{\mathbf{k}}(M \otimes_{\mathbf{k}} L, N) \cong \mathrm{Hom}_{\mathbf{k}}(M, \mathrm{Hom}_{\mathbf{k}}(L, N)) \cong \mathrm{Hom}_{\mathbf{k}}(L, \mathrm{Hom}_{\mathbf{k}}(M, N))$$

So for $f \in \mathrm{Hom}_{\mathbf{k}}(L, R)$ define

$$\theta(f) : \mathrm{Hom}_{\mathbf{k}}(R, \mathrm{Hom}_{\mathbf{k}}(M, N)) \xrightarrow{\circ f} \mathrm{Hom}_{\mathbf{k}}(L, \mathrm{Hom}_{\mathbf{k}}(M, N)).$$

Thus θ is a contravariant functor. Now, let us assume that

$$0 \to L' \to L \to L'' \to 0$$

is a short exact sequence. Let us apply functor θ, which is a composition of functor $\mathrm{Hom}_{\mathbf{k}}(\bullet, N)$ and $\mathrm{Hom}_{\mathbf{k}}(M, \bullet)$. Functor $\mathrm{Hom}_{\mathbf{k}}(\bullet, N)$ is left exact contravariantly, thus getting the exact sequence:

$$0 \to \mathrm{Hom}_{\mathbf{k}}(L'', N) \to \mathrm{Hom}_{\mathbf{k}}(L, N) \to \mathrm{Hom}_{\mathbf{k}}(L', N)$$

The functor $\mathrm{Hom}_{\mathbf{k}}(M, \bullet)$ is right adjoint to the functor $(\bullet) \otimes_{\mathbf{k}} M$, thus is left exact, producing the short exact sequence:

$$0 \to \mathrm{Hom}_{\mathbf{k}}(M, \mathrm{Hom}_{\mathbf{k}}(L'', N)) \to \mathrm{Hom}_{\mathbf{k}}(M, \mathrm{Hom}_{\mathbf{k}}(L, N)) \to \mathrm{Hom}_{\mathbf{k}}(M, \mathrm{Hom}_{\mathbf{k}}(L', N)),$$

which is isomorphic to

$$0 \to \mathrm{Hom}_{\mathbf{k}}(M \otimes L'', N) \to \mathrm{Hom}_{\mathbf{k}}(M \otimes L, N) \to \mathrm{Hom}_{\mathbf{k}}(M \otimes L', N).$$

Thus functor θ is left exact contravariantly.

Problem. Give an example of an additive category which admits neither kernels nor cokernels

Solution. Define $\mathscr{C}$ such that $\mathrm{Ob}(\mathscr{C}) = \mathbb{Z}$ and $\mathrm{Mor}(\mathscr{C})$ are defined as:

$$\mathrm{Hom}_{\mathscr{C}}(n, m) = \begin{cases} \mathbb{Z}, & m = n, \\ \{0\}, & m \neq n. \end{cases}$$

Note that $\mathrm{Mor}(\mathscr{C}) \in \underline{\mathrm{Mod}}(\mathbb{Z})$, composition is bilinear, and finite biproduct exist. Thus, $\mathscr{C}$ is indeed an additive category.

Now, do kernel exist? Suppose it exists and call it $\mathrm{Ker}(f)$ for some non-zero $f : m \to m$. Then there should be a morphism $\iota : \mathrm{Ker}(f) \to m$ such that $f \circ \iota = 0$ and ι being universal. However, all morphisms $\mathrm{Ker}(f) \to m$ for $\mathrm{Ker}(f) \neq m$ are zero and $f \circ \mathrm{id}_m = 1 \neq 0$, so kernel does not exist. Similar logic can be applied to show that cokernel does not exist.

Problem. Let $X = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^3, z > 0\}$, $Y = \mathbb{R}^2$ and let $f : X \to Y$ be the map $(x, y, z) \mapsto (x, y)$. Get $G := Rf_*\mathbf{k}_X$. Calculate the stalk G_0 of G at $0 \in \mathbb{R}^2$.

Solution. Recall that:

$$(Rf_*\mathbf{k}_X)_0 = \varinjlim_{U \ni 0} H(f^{-1}(U), \mathbf{k}_X)$$

Now note that $f^{-1}(U) \cong \mathbb{S}^1$ for any open neighbourhood $U \ni 0$, thus:

$$(Rf_*\mathbf{k}_X)_0 = \varinjlim_{U \ni 0} H(\mathbb{S}^1, \mathbf{k}_X) = H(\mathbb{S}^1, \mathbf{k}_X) \cong H(\mathbb{S}^1, \mathbf{k}_{\mathbb{S}^1}) \cong \boxed{\mathbf{k} \oplus \mathbf{k}[-1]}.$$

Problem. Let $X = \mathbb{R}^2$, $Y = \mathbb{R}$ and let $f : X \to Y$ be the projection $(x, y) \mapsto x$.

(a) Let $Z = \{(x, y) \in X : y = x^2\}$. Compute $Rf_*\mathbf{k}_{XZ}$;

(b) Let $S = \{(x, y) \in X : xy = 1\}$. Compute $Rf_*\mathbf{k}_{XS}$.

Solution. *Part (a).* Similarly to the previous problem:

$$(Rf_*\mathbf{k}_{XZ})_x = \varinjlim_{U \ni x} H(f^{-1}(U), \mathbf{k}_{XZ}) \cong \varinjlim_{U \ni x} H(f^{-1}(U) \cap Z, \mathbf{k}_X)$$

Now note that $f^{-1}(U)$ gives the long vertical stripe, thus $f^{-1}(U) \cap Z$ gives the open connected segment on parabola $y = x^2$, which is homeomorphic to Y. Therefore,

$$(Rf_*\mathbf{k}_{XZ})_x \cong H(\mathbb{R}, \mathbf{k}_{\mathbb{R}^2}) \cong \mathbf{k}$$

Part (b). Previous formula applies:

$$(Rf_*\mathbf{k}_{XS})_x \cong \varinjlim_{U \ni x} H(f^{-1}(U) \cap S, \mathbf{k}_X)$$

Now the expression depends on whether $x = 0$ or not. If $x \neq 0$, we get the same result as before since we get an open connected segment on one of the branches of hyperbola:

$$(Rf_*\mathbf{k}_{XS})_x \cong \mathbf{k}, \ x \neq 0.$$

However, when $x = 0$, we get that $f^{-1}(U) \cap S$ gives two branches of hyperbola, both homeomorphic to $\mathbb{R}$. Therefore, $f^{-1}(U) \cap S \cong \mathbb{R} \sqcup \mathbb{R}$ and so our cohomology:

$$(Rf_*\mathbf{k}_{XS})_0 \cong \mathbf{k}^2$$

Therefore, stalkwise, our resultant sheaf is given by:

$$(Rf_*\mathbf{k}_{XS})_x \cong \begin{cases} \mathbf{k}^2, & x = 0, \\ \mathbf{k}, & x \neq 0. \end{cases}$$

Problem. Let X be a real manifold and let S be a closed subset. Give a necessary and sufficient condition in order that $\Gamma_S(C^\infty_X) \neq 0$.

Solution. Recall that $\Gamma_S(C^\infty_X) = \{f \in C^\infty_X : S \subseteq \operatorname{supp}(f)\}$. We claim that necessary and sufficient condition is $\operatorname{int}(S) \neq \emptyset$. Indeed, if $\Gamma_S(C^\infty_X) \neq 0$, then exists some point $s \in S$ such that $f(s) \neq 0$. Thus, there exists some open neighbourhood U of s such that $f(U) \neq 0$, resulting in $\operatorname{int}(S) \neq \emptyset$. The reverse direction is obvious.

Problem. Consider the map $f : X \to Y$ (where $X = \mathbb{R}^2$, $Y = \mathbb{R}^3$) such that $(x, y) \mapsto (x, y, x + y + 1)$. Let $Z = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2\}$. Compute $f^{-1}\mathbf{k}_{YZ}$.

Solution. Recall that $f^{-1}\mathbf{k}_{Y,Z} \cong \mathbf{k}_{Y,f^{-1}(Z)}$. Thus it suffices to compute $f^{-1}(Z)$. Now suppose $(a, b) \in f^{-1}(Z)$, then $f(a, b) \in Z$ and thus:

$$a^2 + b^2 = (a + b + 1)^2 \implies 2ab + 2a + 2b + 1 = 0$$

Thus $f^{-1}(Z)$ is exactly the hyperbola $H := \{(x, y) \in \mathbb{R}^2 : 2xy + 2x + 2y + 1 = 0\}$ and so $f^{-1}\mathbf{k}_{Y,Z} \cong \mathbf{k}_{YH}$.

Problem. Consider the map $f : X \to Y$ (where $X = \mathbb{R}^2$, $Y = \mathbb{R}$) given by $(x, y) \mapsto y$. Let $Z = \{(x, y) \in \mathbb{R}^2 : y = x^2\}$. Describe the sheaf $Rf_*\mathbf{k}_{YZ}$.

Solution. As we used previously,

$$(Rf_*\mathbf{k}_{XS})_x = \varinjlim_{U \ni x} H(f^{-1}(U) \cap S, \mathbf{k}_X).$$

Now, $f^{-1}(U)$ gives the horizontal stripe. Note that for $x = x^- < 0$, there is no intersection with S, so $(Rf_*\mathbf{k}_{XS})_{x^-} \cong 0$. For $x = 0$, there is a single connected component, so $(Rf_*\mathbf{k}_{XS})_0 \cong \mathbf{k}$. Finally, for $x = x^+ > 0$, we get disjoint union of two contractible sets, thus yielding $(Rf_*\mathbf{k}_{XS})_{x^+} \cong \mathbf{k}^2$. All in all:

$$(Rf_*\mathbf{k}_{XS})_t \cong \begin{cases} \mathbf{k}^2 & t > 0, \\ \mathbf{k} & t = 0, \\ 0 & t < 0. \end{cases}$$

Problem. Let V be a real finite dimensional vector space and let V^* be its dual. One denotes by q_1 and q_2 the projections $V \times V^* \to V$ and $V \times V^* \to V^*$, respectively. Consider the closed set

$$P = \{(x, y) \in V \times V^* : \langle x, y \rangle \geq 0\}.$$

For $G \in \operatorname{Mod}(\mathbf{k}_V)$, one defines the Fourier transform as

$$\mathscr{F}(G) = Rq_{2!}((q_1^{-1}G)_P).$$

Let $\gamma \subseteq V$ be a closed convex proper cone. Calculate $\mathscr{F}(\mathbf{k}_{V\gamma})$.

Solution. Gave up.

Problem. Let X be a topological space. One says that $F \in \mathrm{Mod}(\mathbf{k}_X)$ satisfies the unique continuation principle (UCP, for short) if for any non-empty connected open set $U \subseteq X$ and any closed subset $Z \subseteq U$ such that $Z \neq U$, one has $\Gamma_Z(U; F) = 0$.

(i) Prove that if F satisfies UCP, then so does $F|_M$ and $\Gamma_M(F)$ for any closed subset M of X.

(ii) Prove that if X is open in $\mathbb{R}^n$ and F is soft, then F does not satisfy the UCP.

(iii) Give an example of a sheaf F which satisfies the UCP principle but such that $R\Gamma_Z(U; F) \neq 0$ for some closed subset Z with $\mathrm{Int}(Z) = \emptyset$.

(iv) Prove that if X is open in $\mathbb{R}^n$ and F is soft, then F does not satisfy the UCP.

Problem. Let $M \in \mathrm{Mod}(\mathbf{k})$. Show that, in general, the functor $\mathrm{Hom}_{\mathbf{k}}(M, \bullet)$ does not commute with directed inductive limits.

Solution.

Problem. Assume $\mathbf{k}$ is a field of characteristic 0 and denote by W_n Weyl algebra in n variables over $\mathbf{k}$. We set $O_n = W_n/(W_n\partial_1 + \cdots + W_n\partial_n)$ and recall that $O_n \cong \mathbf{k}[x_1, \ldots, x_n]$ as a left W_n-module. We also set $B_n = W_n/(W_n\partial_1^2 + \cdots + W_n\partial_n^2)$.

(i) First assume $n = 1$ and compute $W_1/(W_1\partial_1^2)$ and the groups $\mathrm{Ext}^j_{W_1}(B_1, O_1)$.

(ii) Prove that there exists an isomorphism of W_n-modules of B_n with a finite direct sums of copies of O_1.

(iii) Calculate the groups $\mathrm{Ext}^j_{W_n}(B_n, O_n)$.

Solution. *Part (i).* So recall that $W_1 = \mathbf{k}[x, \partial]/(\partial x - x\partial - 1)$. Now,

$$W_1/(W_1\partial^2) \cong \mathbf{k}[x, \partial]/(\partial x - x\partial - 1, W_1\partial^2),$$

which is a $\mathbf{k}$-vector space with basis $\{x^i, x^i\partial : i \geq 0\}$. We find the projective resolution of B_1:

$$0 \to W_1 \xrightarrow{\cdot\partial^2} W_1 \to B_1 \to 0$$

Now we apply $\mathrm{Hom}_{W_1}(\bullet, O_1)$ functor to the projective resolution:

$$0 \to \mathrm{Hom}_{W_1}(W_1, O_1) \xrightarrow{\cdot\partial^2} \mathrm{Hom}_{W_1}(W_1, O_1) \to 0$$

But notice that $\mathrm{Hom}_{W_1}(W_1, O_1) \cong O_1$, so

$$0 \to O_1 \xrightarrow{\cdot\partial^2} O_1 \to 0$$

Compute cohomologies (other are zeros):

$$\mathrm{Ext}^0_{W_1}(B_1, O_1) \cong \mathrm{Ker}(O_1 \xrightarrow{\cdot\partial^2} O_1), \quad \mathrm{Ext}^1_{W_1}(B_1, O_1) \cong \mathrm{Coker}(O_1 \xrightarrow{\cdot\partial^2} O_1)$$

Now, note that $\mathrm{Ker}(\cdot)$ is exactly those polynomials $f \in O_1$ such that $\partial^2 f = 0$, thus $\mathrm{Ext}^0_{W_1}(B_1, O_1) \cong \mathbf{k} \oplus \mathbf{k}x$. Trivially $\mathrm{Ext}^1_{W_1}(B_1, O_1) \cong O_1$.

Part (ii). Essentially we have to compute:

$$W_n/(W_n\partial_1^2 + \cdots + W_n\partial_n^2) \cong \frac{\mathbf{k}[x_1, \ldots, x_n, \partial_1, \ldots, \partial_n]}{(x_i\partial_j - x_j\partial_i - \delta_{ij}, W_n\partial_1^2 + \cdots + W_n\partial_n^2)}$$

Now, initially W_n is a infinite-dimensional vector space with basis

$$\{x_1^{\alpha_1} \dots x_n^{\alpha_n} \partial_1^{\beta_1} \dots \partial_n^{\beta_n} : \alpha_i, \beta_i > 0\}$$

And we "kill" all $\partial_i^{\alpha_i}$ for which $\alpha_i \geq 2$. Therefore, $W_n/(W_n\partial_1^2 + \dots + W_n\partial_n^2)$ is a vector space spanned by the basis $\{x_1^{\alpha_1} \dots x_n^{\alpha_n} \partial_1^{b_1} \dots \partial_n^{b_n} : b_i \in \{0,1\}, \alpha_i \geq 0\}$. Now, define the map:

$$\sum_{b \in \{0,1\}^n} \partial_1^{b_1} \dots \partial_n^{b_n} f_{b_1,\dots,b_n}(x) \mapsto (f_{b_1,\dots,b_n})_{b \in \{0,1\}^n}$$

Thus, $B_n = W_n/(W_n\partial_1^2 + \dots + W_n\partial_n^2) \cong O_n^{\oplus 2^n}$.
Part (iii). Note that we have:

$$\mathrm{Ext}^j_{W_n}(B_n, O_n) \cong \mathrm{Ext}^j_{W_n}(O_n^{\oplus 2^n}, O_n) \cong \prod_{2^n} \mathrm{Ext}^j_{W_n}(O_n, O_n)$$

Apply Künneth formula:

$$\mathrm{Ext}^*_{W_n}(O_n, O_n) \cong \bigotimes_{i=1}^{n} \mathrm{Ext}^*_{W_1}(O_1, O_1).$$

Since $\mathrm{Ext}^*_{W_1}(O_1, O_1) \cong \mathbf{k}[0]$, we obtain:

$$\mathrm{Ext}^j_{W_n}(B_n, O_n) \cong \mathbf{k}^{2^n}$$

Note. General Künneth formula:

$$\mathrm{Ext}^n_{R \otimes S}(M \otimes N, M' \otimes N') \cong \bigoplus_{p+q=n} (\mathrm{Ext}^p_R(M, M') \otimes_{\mathbf{k}} \mathrm{Ext}^q_S(N, N'))$$

Problem. We assume that $\mathbf{k}$ is a field of characteristic 0 and we denote by W_n the Weyl algebra in n variables.

(i) Let $I = W_n \cdot x_1^2 + \dots + W_n \cdot x_n^2$ be the left ideal generated by $(x_1^2, \dots, x_n^2)$. Show that a suitable Koszul complex gives a finite free resolution of W_n/I.

(ii) Calculate $\mathrm{RHom}_{W_n}(W_n/I, W_n/I)$.

Solution. *Part (i).* We build Koszul complex $K^\bullet(W_n/I, (x_1^2, \dots, x_n^2))$. We get:

$$0 \to M^{(0)} \xrightarrow{d^0} \dots \xrightarrow{d^{n-1}} M^{(n)} \to 0,$$

Let us see what each $M^{(j)}$ and d^j stands for. Let $(e_1, \dots, e_n)$ be the canonical basis of $\mathbf{k}^n$, denote $e_I = e_{i_1} \wedge \dots \wedge e_{i_j}$ for $I = \{i_1, \dots, i_j\} \subseteq \{1, \dots, n\}$. Then, $M^{(j)} = W_n \otimes \Lambda^j \mathbf{k}^n \cong W_n^{\oplus C_n^j}$. Now, the differential map is defined as:

$$d^j(w \otimes e_I) = \sum_{i=1}^{n} w x_i^2 \otimes e_i \wedge e_I.$$

Note that elements $(x_1^2, \dots, x_n^2)$ form a regular sequence since each is a non-zero divisor of W_n and W_n is a domain. Now, apply the contravariant functor $\mathrm{Hom}(\bullet, W_n/I)$:

$$0 \to \mathrm{Hom}(M^{(n)}, W_n/I) \xrightarrow{d_0^*} \dots \xrightarrow{d_{n-1}^*} \mathrm{Hom}(M^{(0)}, W_n/I) \to 0$$

Now note that $M^{(j)} \cong W_n^{\oplus\binom{n}{j}}$, so $\mathrm{Hom}_{W_n}(M^{(n)}, W_n/I) \cong \mathrm{Hom}_{W_n}(W_n^{\oplus\binom{n}{j}}, W_n/I) \cong (W_n/I)^{\oplus\binom{n}{j}}$. Thus we get:

$$0 \to (W_n/I)^{\oplus\binom{n}{1}} \xrightarrow{d_0^*} \dots \xrightarrow{d_{n-1}^*} (W_n/I)^{\oplus\binom{n}{n}} \to 0,$$

where $d_j^*(f) = f \circ d^j$. One notices:

$$\mathrm{RHom}_{W_n}(W_n/I, W_n/I) \cong \bigoplus_{j=0}^{n} (W_n/I)^{\binom{n}{j}}[-j]$$

Problem. Let P be a polynomial with coefficients in $\mathbb{C}$ of degree $m > 1$, thus defining a continuous map from $X = \mathbb{C}$ to $Y = \mathbb{C}$. Set $S = \{z \in X : P'(z) = 0\}$ and $Z = P(S) \subseteq Y$. Recall that the restriction to $Y \setminus Z$ of the sheaf $P_*(\mathbf{k}_X)$ is a locally constant sheaf of rank m.

(i) Let $P(z) = z^2$. Prove that the sheaf $P_*(\mathbf{k}_X)$ is locally constant on $Y \setminus \{0\}$, is not constant on $Y \setminus \{0\}$ and is not locally constant on Y.

(ii) Let $P(z) = z^2(z-1)^2$. Find the points of Y in a neighbourhood of which the sheaf $P_*(\mathbf{k}_X)$ is not locally constant.

Proof. *Part (i).* Recall $P_*\mathbf{k}_X(U) = \mathbf{k}_X(P^t(U))$. So if U does not contain critical points on P, $P^t(U)$ splits into m disjoint connected components and so $P_*\mathbf{k}_X(U) \cong \mathbf{k}^m$.

Apply this to $P(z) = z^2$. The critical point is $z = 0$, so over $Y \setminus \{0\}$, $P_*\mathbf{k}_X|_{Y\setminus\{0\}}(U) \cong \mathbf{k}^2$, thus locally constant. Let us show that it is not constant. Consider

$$\Gamma(\mathbb{R}_{>0}, P_*\mathbf{k}_X) = \varinjlim_{U \subseteq \mathbb{R}_{>0}} \Gamma(P^{-1}(U), \mathbf{k}_X) \cong \mathbf{k},$$

which is not isomorphic to $\mathbf{k}^2$, as was expected. Now consider the whole Y. Let us compute the stalk at 0:

$$(P_*\mathbf{k}_X)_0 = \varinjlim_{U \ni 0} \Gamma(P^{-1}(U), \mathbf{k}_X) \cong \mathbf{k}$$

Which differs from any stalk $w \neq 0$, where $(P_*\mathbf{k}_X)_w \cong \mathbf{k}^2$. Thus, $P_*\mathbf{k}_X(U)$ is not locally constant over whole Y.

Part (ii). As was previously noted, $P_*(\mathbf{k}_X)|_{Y\setminus Z}$ is locally constant of rank 4 for $S = \{z \in X : P'(z) = 0\}$ and $Z = P(S)$. Compute the derivative:

$$P'(z) = 2z(z-1)^2 + 2z^2(z-1) = 2z(z-1)(2z-1)$$

Thus $Z = P(\{0, 1, \frac{1}{2}\}) = \{0, 1/16\}$.

Problem. Let V be an infinite dimensional $\mathbf{k}$-vector space. Denote by I the set of finite dimensional vector subspaces of V. For sake of clarity, we denote by V_i the vector space associated to $i \in I$. We order I by the relation $i \leq j$ if $V_i \subseteq V_j$.

(i) Prove that I is directed.

(ii) Compute $\varinjlim_{i \in I} V/V_i$.

Part (i). We need to show (i) I is non-empty, (ii) for any i and j in I, there exists $k \in I$ and morphisms $i \to k, j \to k$, and (iii) for any parallel morphisms $f, g : i \rightrightarrows j$, there exists a morphism $h : j \to k$ such that $h \circ f = h \circ g$.

(i) is obvious: any vector space has a finite dimensional subspace. For (ii), assume V_i and V_j are two finite vector spaces. Then, there is always a finite vector space $V_i + V_j$ with trivial maps $V_i \to V_i + V_j$ and $V_j \to V_i + V_j$. Obviously, $V_i + V_j$ is finite dimensional, thus lies in I. (iii) follows from I having an inclusion order.

Part (ii). By the definition of a colimit,

$$\varinjlim_{i\in I} V/V_i = \left(\bigsqcup_{i\in I} V/V_i\right) \Big/ \sim,$$

where $v + V_i \sim w + V_j$ iff exists $V_k \supseteq V_i, V_j$ such that $v + V_k = w + V_k$ in V/V_k. But this obviously always holds, thus $\varinjlim_{i\in I} V/V_i = 0$.

Problem. Let $C_c^\infty(\mathbb{R}^n)$ denotes the $\mathbb{C}$-vector space of $\mathbb{C}$-valued C^∞-functions on $\mathbb{R}^n$ with compact support.

(i) Let $\theta \in C_c^\infty(\mathbb{R})$ with $\int_{\mathbb{R}} \theta(t)dt = 1$. Prove that any $\varphi \in C_c^\infty(\mathbb{R})$ may be uniquely decomposed as $\varphi = \varphi_0 + \lambda\theta$ with $\varphi_0 \in C_c^\infty(\mathbb{R})$, $\lambda \in \mathbb{R}$, and $\int_{\mathbb{R}} \varphi_0(t)dt = 0$. Also prove that there exists $\psi \in C_c^\infty(\mathbb{R})$ such that $\frac{\partial}{\partial x_1}\psi = \varphi_0$.

(ii) Consider the operator $\frac{\partial}{\partial t} : C_c^\infty(\mathbb{R}) \to C_c^\infty(\mathbb{R})$. Prove that its kernel is 0 and its cokernel has a dimension 1.

(iii) We write $x = (x_1, x') \in \mathbb{R}^n$, $x' = (x_2, \ldots, x_n)$. Prove that any $\varphi \in C_c^\infty(\mathbb{R}^n)$ may be uniquely decomposed as $\varphi(x) = \partial_{x_1}\psi + \lambda(x')\theta(x_1)$ with $\psi \in C_c^\infty(\mathbb{R}^n)$ and $\lambda(x') \in C_c^\infty(\mathbb{R}^{n-1})$.

(iv) Consider the complex

$$0 \to \Omega_c^0(\mathbb{R}^n) \xrightarrow{d} \ldots \xrightarrow{d} \Omega_c^n(\mathbb{R}^n) \to 0,$$

where $\Omega_c^p(\mathbb{R}^n)$ denotes the space of p-differential forms on $\mathbb{R}^n$ with C^∞-coefficients and compact support. Show that this complex is a Koszul complex and compute its cohomology.

Part (i). By applying $\int_{\mathbb{R}}(\cdot)$ to both parts of the equation we immediately get $\lambda = \int_{\mathbb{R}} \varphi(t)dt$. Define $\varphi_0 := \varphi - \lambda\theta$. Its integral is obviously 0 and it is in $C_c^\infty(\mathbb{R})$. Uniqueness is obvious.

On the second part, we can pick $\psi(x_1) := \int_{-\infty}^{x_1} \varphi_0(t)dt$. It is clearly in $C^\infty(\mathbb{R})$, so we shall only show that ψ vanishes outside some $[-\rho, \rho]$. Since $\varphi_0 \in C_c^\infty(\mathbb{R})$, it vanishes outside some $[-\gamma, \gamma]$, thus ψ vanishes over $(-\infty, -\gamma]$. Now, since $\int_{\mathbb{R}} \varphi_0(t)dt = 0$, thus for any $r > \gamma$, we have:

$$0 = \psi(\infty) = \psi(r) + \int_r^\infty \varphi_0(t)dt,$$

but the latter term is clearly 0, thus $\psi(r) = 0$. Thus, ψ vanishes outside $[-\gamma, \gamma]$.

Part (ii). Kernel of this map is any $f \in C_c^\infty(\mathbb{R})$ such that $\partial_t f = 0$, meaning f is a constant function. However, constant functions are not in $C_c^\infty(\mathbb{R})$ (except for 0), thus the kernel is 0.

Cokernel is, by definition, $C_c^\infty(\mathbb{R})/\mathrm{im}(\partial_t)$. So what exactly is the image of ∂_t? $g \in \mathrm{im}(\partial_t)$ if $g = \partial_t f$ for some $f \in C_c^\infty(\mathbb{R})$. Clearly if $\int_{\mathbb{R}} g(t)dt = 0$, then $g \in \mathrm{im}(\partial_t)$ by *part (i)*. In general, it is easy to show that if $\int_{\mathbb{R}} g(t)dt \neq 0$, then there are no functions $h \in C_c^\infty(\mathbb{R})$ such that $\partial_t h = g$. Therefore, $\mathrm{im}(\partial_t) = \{f \in C_c^\infty(\mathbb{R}) : \int_{\mathbb{R}} f(t)dt = 0\}$ and thus $C_c^\infty(\mathbb{R})/\mathrm{im}(\partial_t)$ gives equivalence

classes where $f \sim g$ iff $\int_{\mathbb{R}} f(t)dt = \int_{\mathbb{R}} g(t)dt$. But notice that the equivalence class is fully determined by the value of the integral $\int_{\mathbb{R}} f(t)dt$. Thus, $C_c^\infty(\mathbb{R})/\mathrm{im}(\partial_t) \cong \mathbb{C}$ is of rank 1.

Part (iii). Apply $\int_{\mathbb{R}}(\bullet)(t,x')dt$ to both sides. We get $\int_{\mathbb{R}} \varphi(t,x')dt = \lambda(x')$. Thus, define ψ such that $\partial_{x_1}\psi = \varphi(x) - \lambda(x')\theta(x_1)$. Notice that $\varphi(x) - \lambda(x')\theta(x_1) \in C_c^\infty(\mathbb{R}^n)$ automatically. Thus our proof boiled down to showing:

Claim. For any $f \in C_c^\infty(\mathbb{R}^n)$ such that $\int_{\mathbb{R}} f(t,x')dt = 0$, there exists a unique $g \in C_c^\infty(\mathbb{R}^n)$ such that $f = \partial_{x_1} g$.

This is easily checked by induction for $g(x) = \int_{-\infty}^{x_1} f(t,x')dt$.

Part (iv). Note that $\partial = (\partial_{x_1}, \ldots, \partial_{x_n})$ are trivially $\mathbb{C}$-linear endomorphisms of $C_\infty^c(\mathbb{R}^n)$ which commite with one another: $[\partial_{x_i}, \partial_{x_j}] = 0$. Thus, from $\Omega^p(\mathbb{R}^n)$ being a Koszul complex, we immediately get that this is a Koszul complex $K^\bullet(C_c^\infty(X), \partial)$.

Problem. Let $\mathscr{C}$ be an additive category.

(i) Let $u : X_0 \to Y_0$ and $v : X_1 \to Y_1$ be two morphisms in $\mathscr{C}$. Prove that if $u \oplus v : X_0 \oplus X_1 \to Y_0 \oplus Y_1$ is an isomorphism, then both u and v are isomorphisms.

(ii) Give an example of an isomorphism $w : X_0 \oplus X_1 \to Y_0 \oplus Y_1$ for which X_0 and Y_0 are not isomorphic.

Solution. *Part (i).* Since $u \oplus v = \begin{bmatrix} u & 0 \\ 0 & v \end{bmatrix}$ is an isomorphism, these exists some matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $a, c : Y_0 \to X_0$ and $b, d : Y_1 \to X_1$ such that:

$$\begin{bmatrix} u & 0 \\ 0 & v \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \mathrm{id}_{Y_0} & 0 \\ 0 & \mathrm{id}_{Y_1} \end{bmatrix}, \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} u & 0 \\ 0 & v \end{bmatrix} = \begin{bmatrix} \mathrm{id}_{X_0} & 0 \\ 0 & \mathrm{id}_{X_1} \end{bmatrix}.$$

From here one deduces $b = c = 0$ and:

$$u \circ a = \mathrm{id}_{Y_0}, \quad a \circ u = \mathrm{id}_{X_0}, \quad v \circ d = \mathrm{id}_{Y_1}, \quad d \circ v = \mathrm{id}_{Y_0},$$

which naturally induce an isomorphism with $u^{-1} = a$, $v^{-1} = d$.

Part (ii). Take $w : X \oplus Y \to Y \oplus X$ induced by matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

Problem. Let $\mathbf{k}$ be a field and denote by $\mathrm{Mod}^f(\mathbf{k})$ the abelian category of finite dimensional $\mathbf{k}$-vector spaces. For $V^\bullet \in C^b(\mathrm{Mod}^f(\mathbf{k}))$ one defines its Euler-Poincaré index $\chi(V)$ by the formula:

$$\chi(V) = \sum_i (-1)^i \dim H^i(V).$$

(i) Prove that if the complex V is exact, then $\chi(V) = 0$.

(ii) Let $0 \to V' \to V \to V'' \to 0$ be an exact sequence in $C^b(\mathrm{Mod}^f(\mathbf{k}))$. Prove that $\chi(V) = \chi(V') + \chi(V'')$.

(iii) Let $V, W \in C^b(\mathrm{Mod}^f(\mathbf{k}))$. Prove that $\chi(V \otimes W) = \chi(V)\chi(W)$.

Solution. *Part (i).* By the definition of exactness, $\mathrm{im}(d_V^{i-1}) \cong \ker(d_V^i)$, thus cohomology groups are $H^i(V) = \ker(d_V^i)/\mathrm{im}(d_V^{i-1}) \cong 0$. Trivially, this implies $\chi(V) = 0$.

Part (ii). A short exact sequence of complexes induces a long exact sequence in cohomology:

$$\cdots \to H^i(V') \to H^i(V) \to H^i(V'') \xrightarrow{\delta^i} H^{i+1}(V') \to \ldots$$

Set $a_i := \dim \operatorname{Im}(H^i(V') \to H^i(V))$, $b_i := \dim \operatorname{Im}(H^i(V) \to H^i(V''))$ and $c_i := \dim \operatorname{Im}(\delta^i)$. Then:

$$\begin{aligned} \dim H^i(V') &= a_{i-1} + c_{i-1}, \\ \dim H^i(V) &= a_i + b_i, \\ \dim H^i(V'') &= b_i + c_i. \end{aligned}$$

One then easily checks that $\chi(V) - \chi(V') - \chi(V'') = \sum_i (-1)^i (a_{i-1} - a_i + c_{i-1} - c_i) = 0$.

Part (iii). We apply the Künneth formula:

$$H^n(V \otimes W) \cong \bigoplus_{p+q=n} H^p(V) \otimes H^q(W).$$

Note that $\dim(H^p(V) \otimes H^q(W)) = \dim H^p(V) \dim H^q(W)$, thus:

$$\dim H^n(V \otimes W) = \sum_{p+q=n} \dim H^p(V) \dim H^q(W)$$

Now compute $\chi(V \otimes W)$:

$$\begin{aligned} \chi(V \otimes W) = \sum_i (-1)^i \dim H^i(V \otimes W) &= \sum_i (-1)^i \sum_{p+q=i} \dim H^p(V) \dim H^q(W) \\ &= \sum_i (-1)^i \dim H^i(V) \cdot \sum_j (-1)^j \dim H^j(W) \\ &= \chi(V)\chi(W). \end{aligned}$$

Problem. Let $X = \mathbb{R}$ and let $F = \bigoplus_{n>0} \mathbf{k}_{1/n}$, where $\mathbf{k}_{1/n}$ is the skyscraper sheaf supported by the point $1/n$. Calculate the stalk F_0.

Solution. Note:

$$\Gamma(U, F) = \Gamma\left(U, \bigoplus_{n>0} \mathbf{k}_{1/n}\right) = \bigoplus_{1/n \in U} \mathbf{k}$$

Thus $F_0 = \varinjlim_{0 \ni U} \Gamma(U, F) = \varinjlim_{0 \ni U} \bigoplus_{1/n \in U} \mathbf{k}$. However, we can take U arbitrarily small to make corresponding sections 0, thus $F_0 = 0$.

Problem. Let $\mathscr{C}$ be a category with the property that $X, Y \in \mathscr{C}$ are isomorphic iff $X = Y$. Let $\mathscr{C} \cong \mathscr{C}'$ be an equivalence of categories. Does $\mathscr{C}'$ have the same property as $\mathscr{C}$?

Solution. Let us gather all the facts:

1. $f : X \to X$ is an isomorphism in $\mathscr{C}$ if exists $g : X \to X$ such that $f \circ g = g \circ f = \mathrm{id}_X$.
2. $\mathrm{Hom}_{\mathscr{C}}(X, Y) \cong \mathrm{Hom}_{\mathscr{C}'}(F(X), F(Y))$.
3. For each $Y \in \mathscr{C}'$ exists $X \in \mathscr{C}$ and $F(X) \cong Y$.

Notice condition 3: the inverse of the given $Y \in \mathscr{C}'$ might not be unique. This way, while we have an isomorphism $X \to X$, we might still have isomorphisms of form $X' \to Y'$ in $\mathscr{C}'$ for $X' \neq Y'$.

Suppose $\mathrm{Ob}(\mathscr{C}) = \{\mathbf{k}^1, \mathbf{k}^2, \dots\}$ and maps are linear maps. Take $\mathrm{Ob}(\mathscr{C}') = \mathrm{Vect}(\mathbf{k})$. Take functor $F : \mathbf{k}^n \mapsto \mathbf{k}^n$ that assings to $\mathbf{k}^n$ itself. Note that this is an equivalence: for any $V \in \mathrm{Vect}(\mathbf{k})$ there exists $W \in \mathscr{C}$ (namely, $W = \mathbf{k}^{\dim V}$) such that $F(W) \cong V$. Isomomorphism of Hom's is obvious.

Note that for example $\mathbf{k}[x]_{\leq 1} \cong \mathbf{k}^2$, but they are not equal.

Problem. Let $F : \mathscr{C} \to \mathscr{C}'$ be an additive functor of abelian categories.

(i) Prove that if F is fully faithful, then it is conservative.

(ii) Give an example of a functor which is conservative, but not fully faithful.

(iii) Give an example of a functor which is full but not faithful.

(iv) Give an example of a functor which is faithful but not full.

Solution. *Part (i).* Assume $F(f) \in \mathrm{Hom}(X', Y')$ is an isomorphism between $X', Y' \in \mathscr{C}'$. By essential surjectivity, there exist X, Y such that $F(X) \cong X'$, $F(Y) \cong Y'$. Moreover, since $\mathrm{Hom}_{\mathscr{C}}(X, Y) \cong \mathrm{Hom}_{\mathscr{C}'}(X', Y')$ due to fully faithfulness, there exists $g \in \mathrm{Hom}_{\mathscr{C}'}(X', Y')$ such that $F(f) \cong g$. We claim $g = f$.

Part (ii). Forgetful functor $\mathbf{Ab} \to \mathbf{Set}$.

Part (iii). Any map from non-trivial category to the category with 1 object.

Part (iv). Forgetful functor $\mathbf{Ring} \to \mathbf{Ab}$.